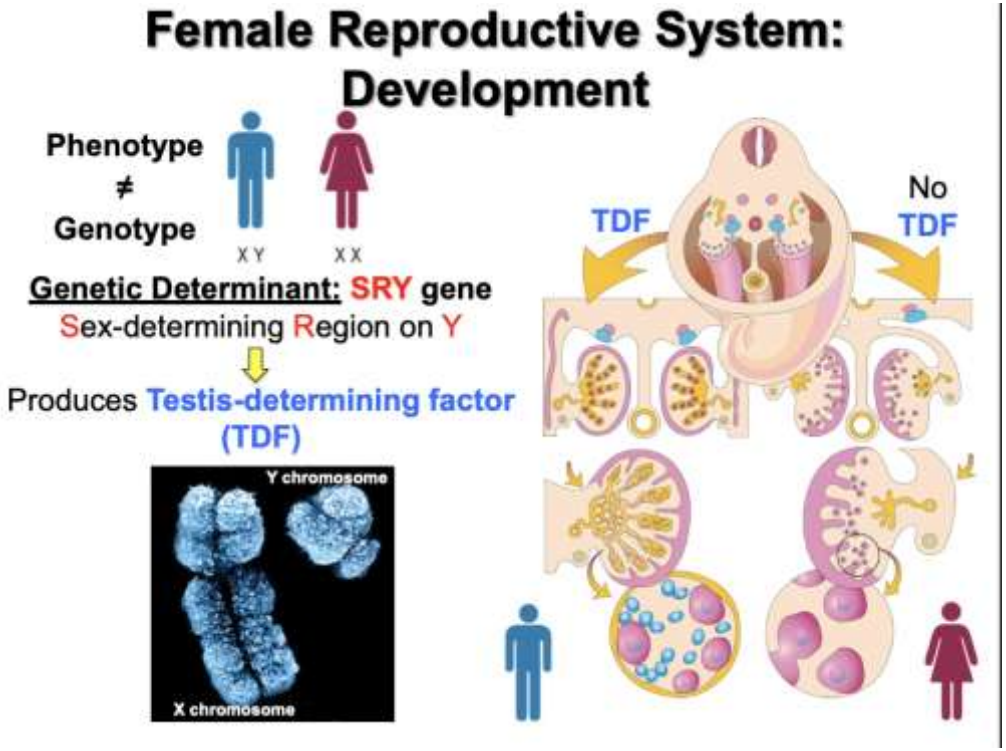
Week #3: [Embryology of the Female Reproductive Tract]

Review of the Indifferent Stage of Sexual Development
Development of the Female Reproductive Tract & External Genitalia

REVIEW of the INDIFFERENT STAGE OF SEXUAL DEVELOPMENT



1. Sexual Differentiation:

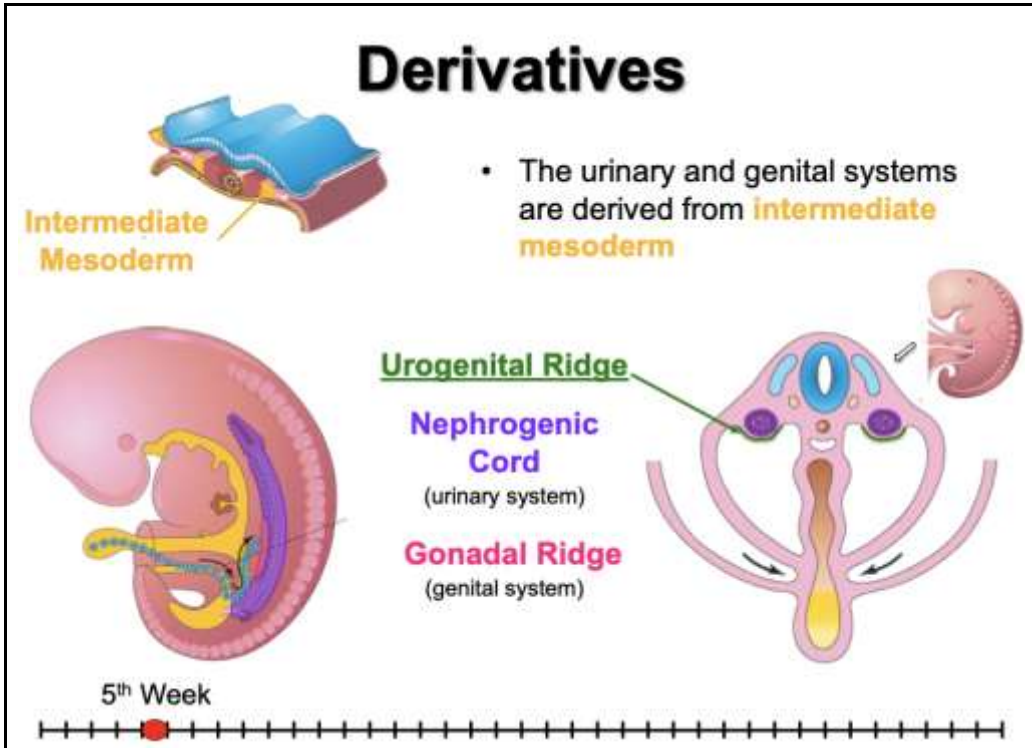
a. **Genetic sex (genotype)** – determined at fertilization; dependent on sex chromosomes
b. **Phenotypic sex (physical gender)** – not always genetic sex; may be dependent on environmental factors (hormonal/tissue interactions)
c. All embryos start out exactly the same until the primordial germ cells arrive

2. Genetic Determinant: SRY (Sex-determining Region on Y) gene is located on the short arm of the Y chromosome; it produces **testes-determining factor (TDF)**, a transcription factor crucial for testes differentiation.

a. If ***present***, male development occurs (acts as a switch to orchestrate a cascade of gene transcriptionsèmale)
b. If ***not present***, female phenotype occurs (default pattern)

3. Timeline for the urogenital system:

a. Development of the urinary system precedes genital development by ~10 days. The UG system is the last system to develop
b. Landmark stages:
   i. **Weeks 4-7**: indifferent stage
   ii. **Week 12:** sexually distinct; definitive kidneys have ascended and are functional
   iii. **Weeks 16-20**: ultrasounds are routinely used to detect phenotypic sex
   iv. **6-9 months:** testes migrate through inguinal canal and enter scrotum

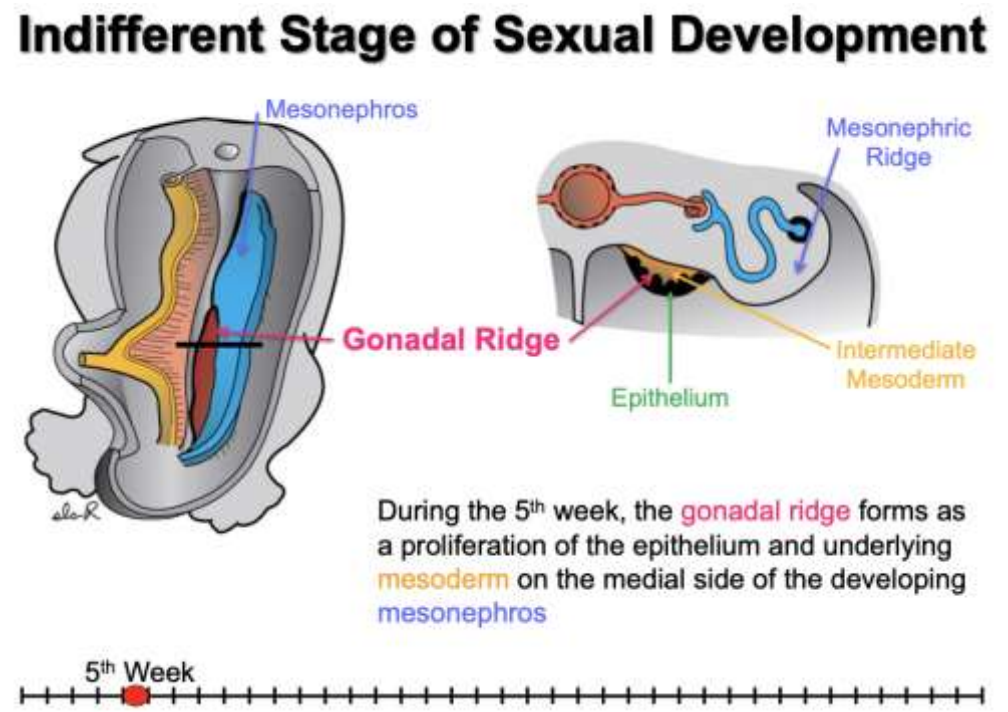


Interconnections between urinary and genital development

a. **Intermediate mesoderm** forms most of the genitourinary system; exceptions:
   i. **Endoderm** from hindgut forms the urogenital sinus
   ii. **Lateral plate mesoderm** forms genital tubercle, lower body wall, and urorectal septum

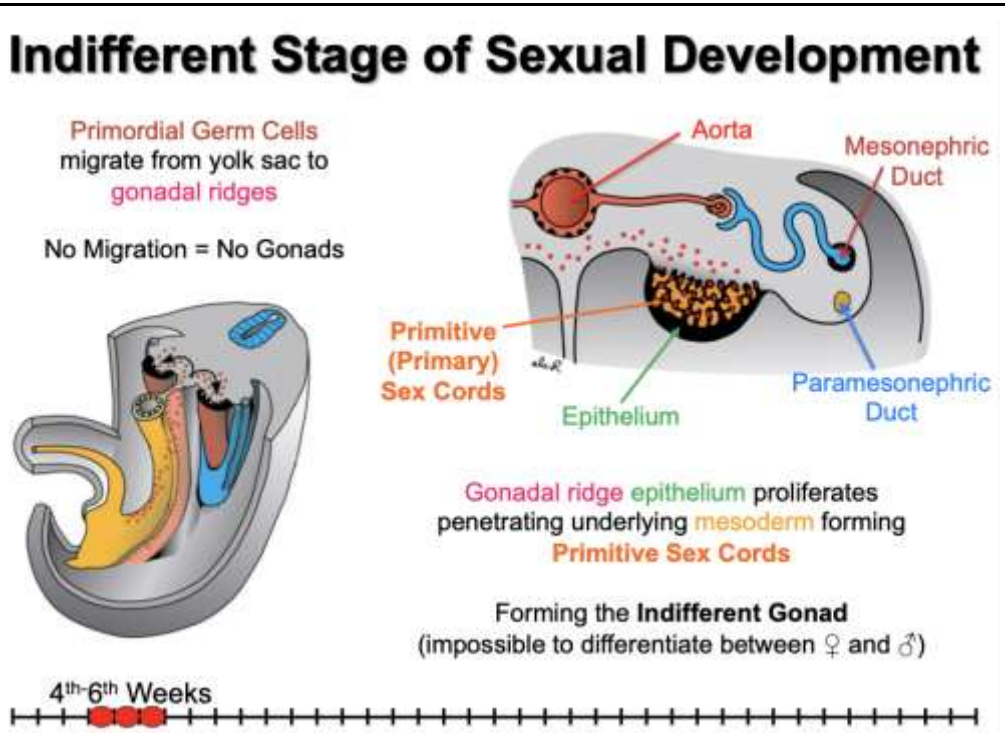
Indifferent Gonads:

a. **Urogenital ridge** – formed from ***intermediate mesoderm*** along posterior body wall on each side of the dorsal aorta; gives rise to the developing gonad and mesonephric kidney
b. During the 5th week, the **gonadal (genital) ridge** forms as a proliferation of **epithelium** and underlying **mesoderm** on the medial side of the developing mesonephros



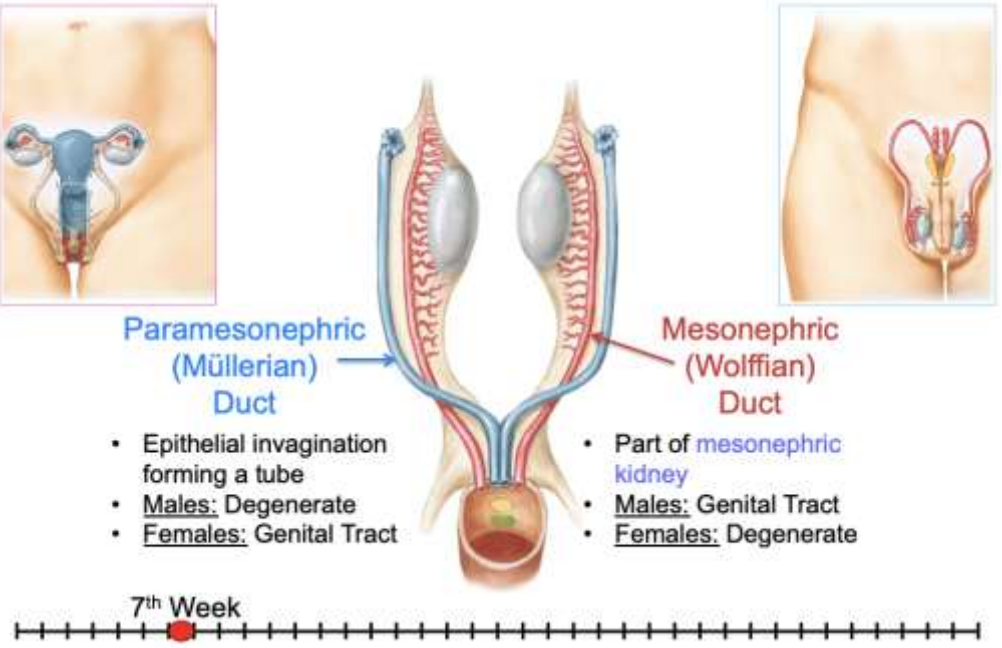
Primordial Germ Cells:

a. Originate in the epiblast, migrate through the primitive streak, and by the 3rd week, reside among endoderm cells in the wall of the **yolk sac** close to the allantois
b. During the 4th-6th weeks, **primordial germ cells** migrate from the neck of the yolk sac along the hindgut mesentery to the mesoderm of the posterior body wall (~T10 vertebral level) and onto the **gonadal ridges**
   i. If the primordial germ cells do not reach the gonadal ridges, the gonads do not develop. Hence, these cells have an inductive influence on development of the gonad into the ovary or testis.



Primitive (Primary) Sex Cords:

a. Shortly before/during arrival of the primordial germ cells, the **genital ridge epithelium proliferates**, and the epithelial cells penetrate into the underlying mesoderm forming fingerlike, epithelial projections called **primitive sex cords**
b. Forms an **indifferent gonad** (impossible to differentiate between male and female)

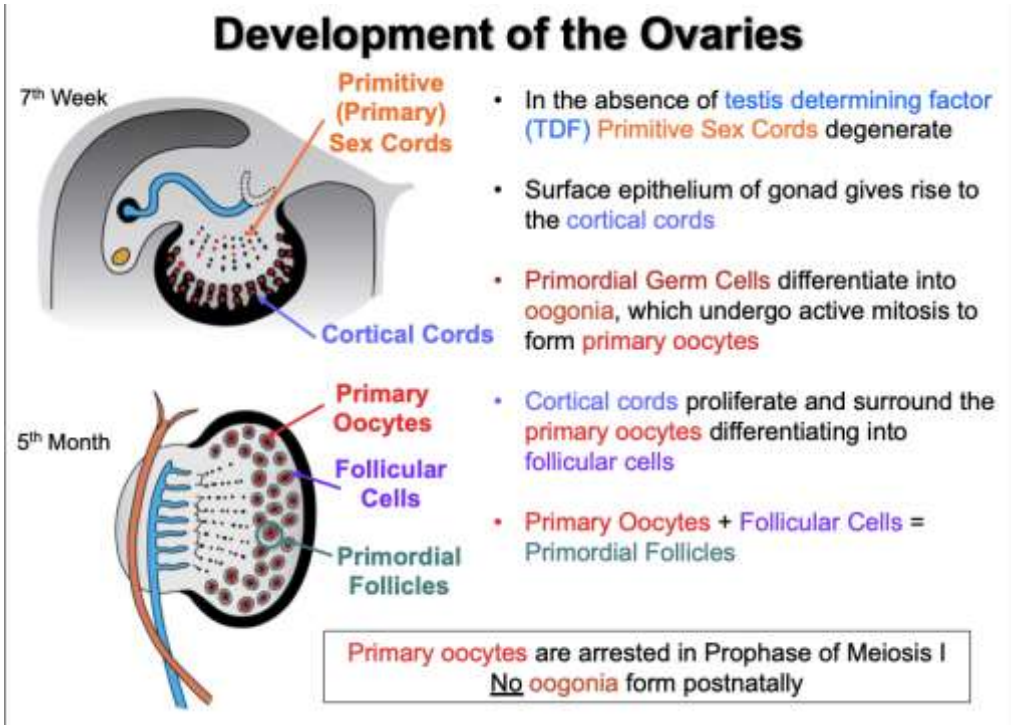


Genital Ducts:

a. Two pairs of genital ducts initially form in both male and female embryos:
    i. **Mesonephric (Wolffian) Ducts**
        1. Develop initially as part of the mesonephric kidney
        2. Undergo extensive lengthening and cell growth to become the genital tract in males; degenerate in females
    ii. Paramesonephric (Müllerian) Ducts
        1. Develop as invaginations of the epithelium on the lateral surface of the urogenital ridge; the edges fuse to form a tube
        2. Undergo further differentiation to form the uterus and a portion of the vagina in females; **degenerate in males**

- The female phenotype develops if TDF, produced by the SRY gene, is NOT present
- The female reproductive tract forms from the paramesonephric duct

DEVELOPMENT of the FEMALE REPRODUCTIVE TRACT & EXTERNAL GENITALIA

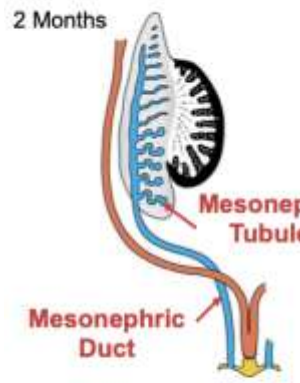


Female Development and Differentiation Patterns of Internal Genitalia

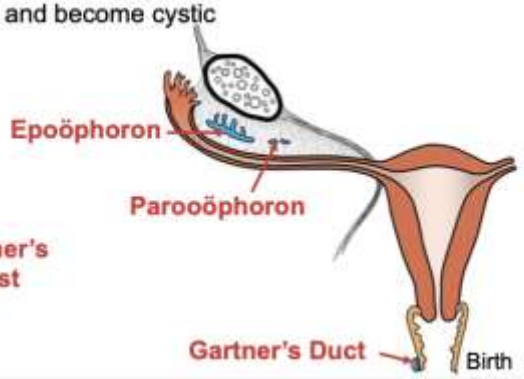
1. **Development of the Ovaries (XX Female)**
   a. **Primitive sex cords** degenerate
   b. **Cortical (secondary) sex cords** develop as a second wave of sex cords as the surface epithelium (cortex) continues to proliferate and invade between the germ cells (week 7)
   c. In the 3rd month, the **cortical cords** split into isolated clusters surrounding the germ cells.
   d. Germ cells develop into **oogonia**. Epithelial cells, derived from the **cortical cords**, surrounding the oogonia differentiate into **follicular cell**. Together a primary oocyte and follicular cells constitute a primordial follicle.
   e. **Oogonia** undergo active mitosis during the fetal period.
   f. **Oogonia** enlarge to form **primary oocytes** that are arrested in **prophase of meiosis I** until shortly before ovulation.
   g. No oogonia form postnatally. At birth, ovaries contain ~400,000 follicles arrested in **prophase of meiosis I**

Fate of the Mesonephric Ducts

- In the absence of testis determining factor (TDF) mesonephric tubules and ducts degenerate
- Some non-functioning remnants remain:
  - Epoöphoron
  - Paroöphoron
  - Gartner's Duct
- Clinically, these structures sometimes accumulate fluid and become cystic





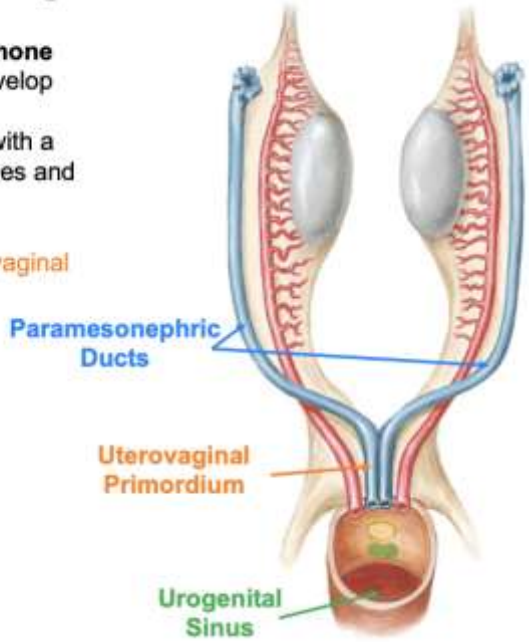


Mesonephric Ducts:

a. In the absence of **testosterone** the mesonephric mesoderm degenerates and the mesonephric tubules and ducts regress
b. Remnants: (found within the broad ligament of the uterus)
   i. **Epoöphoron** – corresponds to the efferent ductules and duct of the epididymis; located within the mesovarium
   ii. **Paroöphoron** – rudimentary mesonephric tubules located closer to the uterus
   iii. **Gartner's duct** – corresponds to the ductus deferens and ejaculatory duct; located along the lateral wall of the uterus within the broad ligament and in the vaginal wall
   iv. **Clinical Note:** These structures sometimes accumulate fluid and become cystic, which may cause problems

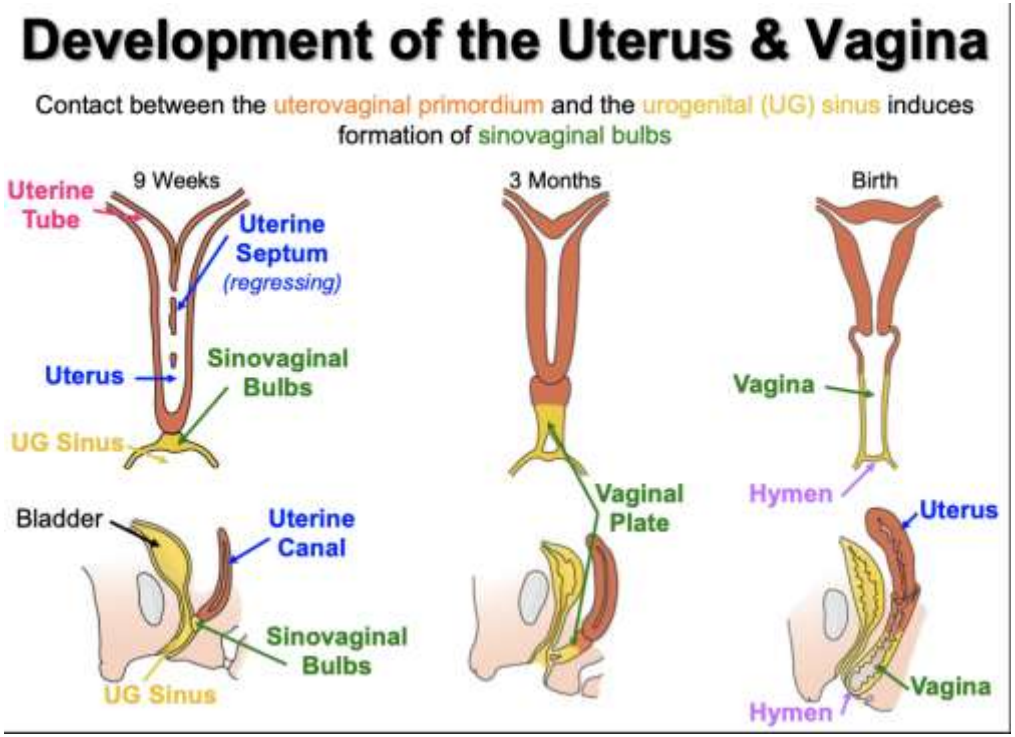
Paramesonephric Ducts

- In the absence of **anti-müllerian hormone (AMH)** the paramesonephric ducts develop
- Cranially, open into abdominal cavity with a funnel-type end to form the uterine tubes and fimbria
- Caudally, ducts fuse to form the uterovaginal primordium
  - Gives rise to:
    - Uterus
    - Cervix
    - Superior 1/3 of Vagina



Paramesonephric Ducts:

a. In the absence of **anti-müllerian hormone (AMH)** paramesonephric ducts develop
b. Cranially, paramesonephric ducts open into the abdominal cavity with a funnel-like end. This portion of the ducts will form the **right** and **left uterine tubes**
c. Caudally, the ducts fuse in the midline to form the uterovaginal primordium, which will form the uterus, cervix, and superior 1/3 of the vagina



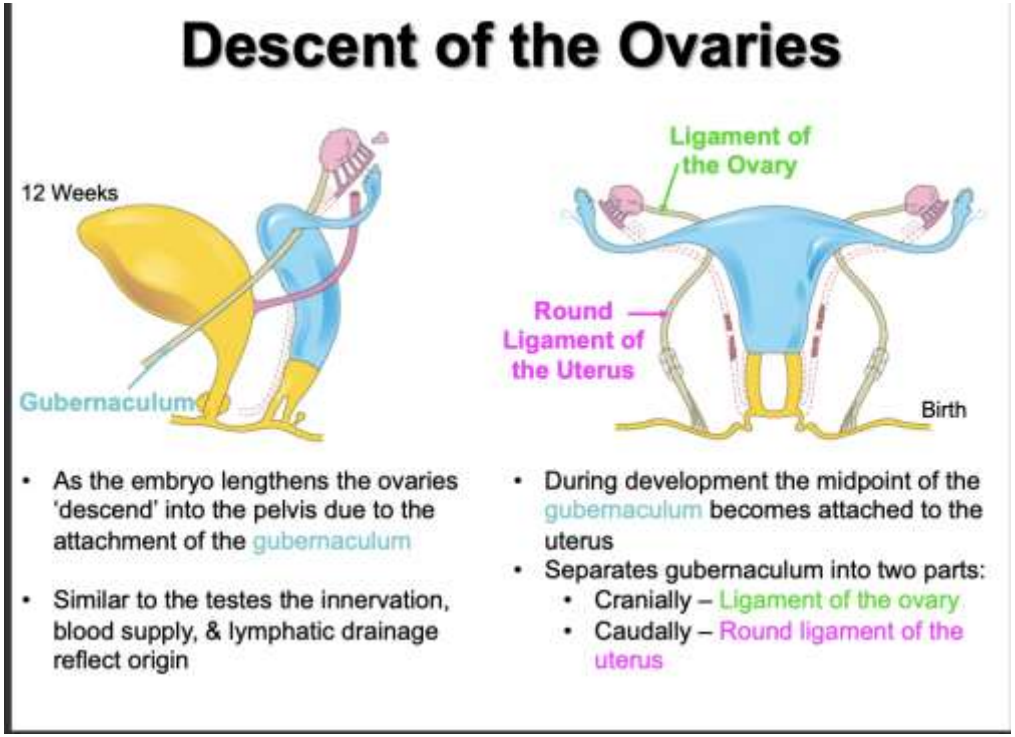
A **bicornuate uterus** develops if the paramesonephric ducts forming the superior part of the body of the uterus fail to fuse. A **unicornuate uterus** develops when one paramesonephric duct fails to develop, resulting in a uterus with uterine tube. In many cases the individuals are fertile but may have an increased incidence of premature delivery.

Development of the Uterus and Vagina:

a. The **uterovaginal primordium** (fused paramesonephric ducts) projects into the dorsal wall of the urogenital (UG) sinus, inducing formation of the ***sinovaginal bulbs*** (paired endodermal outgrowths extending from the UG sinus)
b. **Sinovaginal bulbs** fuse to form the ***solid vaginal plate***, which canalizes and develops into the **inferior 2/3 of the vagina**
c. The lumen of the vagina is separated from the UG sinus cavity by a membranous **hymen** that ruptures during the perinatal period

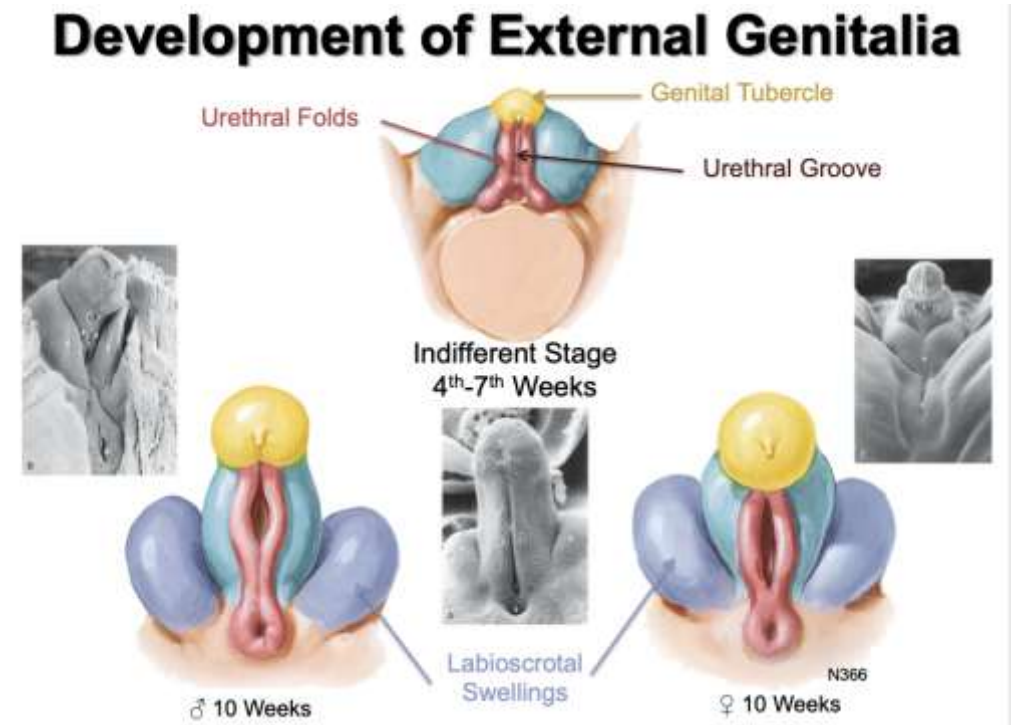
Clinical Note: Failure of the sinovaginal bulbs to develop and form the vaginal plate results in an **absence of the vagina** (1:5000). When the vagina is absent, the uterus is usually absent because the developing uterus (uterovaginal primordium) induces formation of the sinovaginal bulbs, which fuse to form the vaginal plate. Failure of canalization of the vaginal plate results in **vaginal atresia.**

Clinical Note: A double uterus (uterus didelphys) results from failure of fusion of the inferior parts of the paramesonephric ducts. It may be associated with a double or single vagina. A **septate uterus** results from partial or complete failure of resorption of the uterovaginal septum after fusion of the paramesonephric ducts.



Descent of the Ovaries into the Lesser Pelvis

a. As the embryo lengthens and due to the tethering forces of the **gubernaculum**, the ovaries also descend retroperitoneally from the lumbar region of the posterior abdominal wall into the pelvis, but they do not enter the inguinal canal
   i. Innervation, blood supply, and lymphatic drainage all reflect the posterior abdominal origin of the ovaries
b. The midpoint of the gubernaculum becomes attached to the developing uterus near the attachment of the uterine tube
c. The adult derivatives of the female gubernaculum are:
   i. **Ligament of the ovar**y – between the uterus and ovary (cranial portion)
   ii. **Round ligament of the uterus** – connects the uterus to the labia majora (caudal portion)



Development of the External Genitalia:

1. Indifferent Stage:

a. Both sexes look identical from the 4th-7th week. They usually do not acquire distinct masculine or feminine characteristics until the **12th week** (embryonic age).
b. Proliferation of mesoderm around the cloacal membrane results in the formation of:
   i. **Genital tubercle** – located anterior to the UG sinus; it elongates to form a phallus
   ii. **Urethral (urogenital) folds** – form on each side of the cloacal membrane; they form the lateral walls of the urethral groove (on ventral surface).
   iii. **Labioscrotal swellings** – develop on each side of the urogenital folds; primordial scrotum and labia

Development of ♀ External Genitalia

Stimulated by Estrogens & absence of DHT

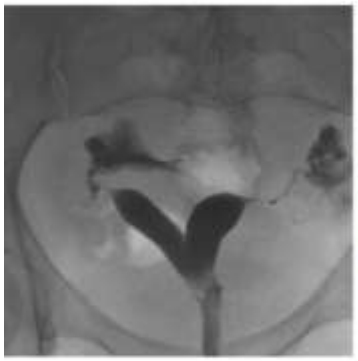


9th-14th Week

Female Differentiation (9-14 weeks)

a. **Estrogens** stimulate development of the external genitalia in the female
b. **Urogenital membrane** breaks down at week 7, exposing the **vestibule of the vagina**
c. **Genital tubercle/phallus** elongates slightly to form the clitoris
d. **Urogenital (urethral) folds** remain separated; develop into the **labia minora**
e. **Labioscrotal swellings** enlarge and remain separated; form the **labia majora**

What Caused These Anomalies?



_____:
- Failure of _____ of the _____





_____:
- Failure of _____ of the _____

_____:
- Each _____ has a _____; may have double _____
- Each _____ forms _____







What Caused These Anomalies?



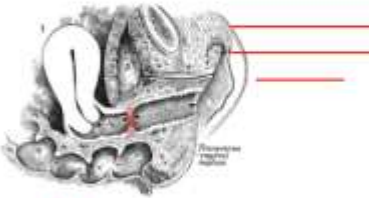
_____:
- Failure of _____ to develop





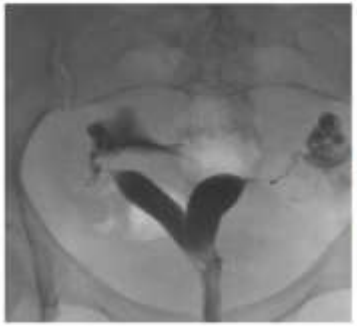
_____:
- A collection of _____ and _____ within the uterus &/or vagina

_____:
- Failure of _____ of the _____ resulting in _____ of the _____
  - _____ – Junction of upper ⅓ and lower ⅔
  - _____ – Caudal end of vagina





What Caused These Anomalies?



Bicornuate Uterus:
- Failure of complete fusion of the paramesonephric ducts




Septate Uterus


Partial Septum

Septate Uterus:
- Failure of degeneration of the uterine septum

Double Uterus (Uterus Didelphys):
- Each uterus has a cervix; may have double vaginas
- Each paramesonephric duct forms independently







What Caused These Anomalies?



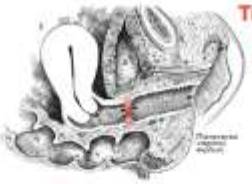
Unicornuate Uterus:
- Failure of one paramesonephric duct to develop





Hematometrocolpos:
- A collection of blood and fluid within the uterus &/or vagina

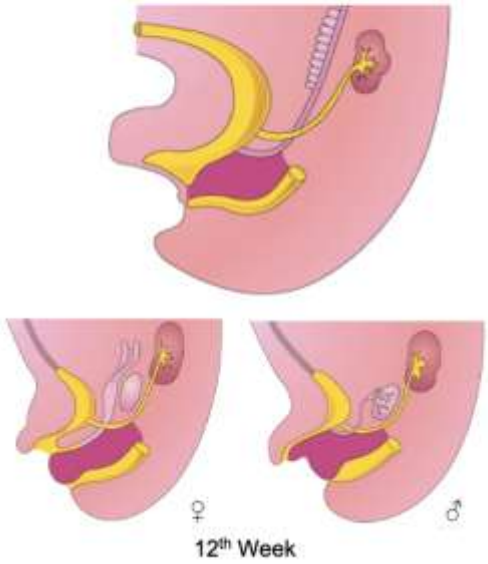
Vaginal Atresia:
- Failure of canalization of the vaginal plate resulting in blockage of the vagina
  - **Transverse Vaginal Septum** –Junction of upper ⅓ and lower ⅔
  - **Imperforate Hymen** – Caudal end of vagina


Transverse Vaginal Septum


Normal

Imperforate

Why are reproductive congenital defects related to Urinary defects?



12th Week

Hematometrocolpos caused by vaginal atresia (congenital)

- Transverse veginal septum usually occurs at the junction between upper ⅓ and lower ⅔ in he vagina
- **Imperforate Hymen** - this is the most common anomaly of the female reproductive tract that results in obstruction